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LESSON PLAN	**COURSE: MHF4U**
Date: Lesson #1	**Unit of Study: Chapter #1 Polynomial Functions**
Materials / Resources: - **Advanced Function 12 Study Guide** - **Chapter #1 & 2 Pre-assessment Quiz** - **Chapter #1 Formula, Aid and Homework Sheet(s)**	**Lesson Focus** - **Power Functions** - **Characteristics of Polynomial Functions** - **Equations and Graphs of Polynomial Functions**

Prior Knowledge:

- What defines a "function"
- Linear functions and their characteristics (e.g., end behaviour and slopes)
- Quadratic functions and their characteristics (e.g., maximum, minimum, and intercepts)
- Able to determine the differences between polynomial, radical, and rational expressions

Overall Expectations:

1. Identify and describe some key features of polynomial functions, and make connections between the numeric, graphical, and algebraic representations of polynomial functions
2. Identify and describe some key features of the graphs of rational functions, and represent rational functions graphically

Specific Expectations:

1.1 Recognize a polynomial expression (i.e., a series of terms where each term is the product of a constant and a power of x with a nonnegative integral exponent, such as $x^3 - 5x^2 + 2x - 1$); recognize the equation of a polynomial function, give reasons why it is a function, and identify linear and quadratic functions as examples of polynomial functions

1.2 Compare, through investigation using graphing technology, the numeric, graphical, and algebraic representations of polynomial (i.e., linear, quadratic, cubic, quartic) functions (e.g., compare finite differences in tables of values; investigate the effect of the degree of a polynomial function on the shape of its graph and the maximum number of x-intercepts; investigate the effect of varying the sign of the leading coefficient on the end behaviour of the function for very large positive or negative x-values)

1.3 Describe key features of the graphs of polynomial functions (e.g., the domain and range, the shape of the graphs, the end behaviour of the functions for very large positive or negative x-values)

1.5 Make connections, through investigation using graphing technology (e.g., dynamic geometry software), between a polynomial function given in factored form [e.g., $fx = 2(x - 3)(x + 2)(x - 1)$] and the x-intercepts of its graph, and sketch the graph of a polynomial function given in factored form using its key features (e.g., by determining intercepts and end behaviour; by locating positive and negative regions using test values between and on either side of the x-intercepts)

1.7 Determine an equation of a polynomial function that satisfies a given set of conditions (e.g., degree of the polynomial, intercepts, points on the function), using methods appropriate to the situation (e.g., using the x-intercepts of the function; using a trial-and-error process with a graphing calculator or graphing software; using finite differences), and recognize that there may be more than one polynomial function that can satisfy a given set of conditions (e.g., an infinite number of polynomial functions satisfy the condition that they have three given x-intercepts)

1.8 Determine the equation of the family of polynomial functions with a given set of zeros and of the member of the family that passes through another given point [e.g., a family of polynomial functions of degree 3 with zeros 5, –3, and –2 is defined by the equation $f(x) = k(x - 5)(x + 3)(x + 2)$, where k is a real number, k ≠ 0; the member of the family that passes through (–1, 24) is $f(x) = - 2(x - 5)(x + 3)(x + 2)$]

1.9 Determine, through investigation, and compare the properties of even and odd polynomial functions [e.g., symmetry about the y-axis or the origin; the power of each term; the number of x-intercepts; $f(x) = f(- x)$ or $f(- x) = - f(x)$], and determine whether a given polynomial function is even, odd, or neither

Learning Goals:

- Students will be able to identify different polynomial functions (e.g., whether they are even or odd, linear, quadratic, cubic, or quartic)
- Students will be able to determine the characteristic of polynomial functions including their domain, range, end behaviour, maximum and minimum
- Students will have knowledge of critical points within a polynomial function and how they are obtained from their equations such as zeros and intercepts
- Students will learn be able to graph polynomial functions from a given polynomial equation
- Distinguish polynomial functions from exponential and sinusoidal functions

Success Criteria:

1. By the end of this lesson I will be able to identify polynomial functions within the power series
2. By the end of this lesson I will be able to determine the characteristics of a polynomial function while observing the equation
3. By the end of this lesson I will be able to calculate the critical points of a polynomial equation, then identify them on a graph
4. By the end of this lesson I will be able to graph polynomial functions

Assessment Strategies:

Diagnostic: chapter #1 & 2 pre-assessment quiz

Formative: homework for chapters 1.1, 1.2 & 1.3

Summative: quiz #1

Time:	**Introduction:**
Part 1	Introduce the students to the course outline and course structure. Students will then be given a pre-assessment quiz to determine their knowledge standings for polynomial functions, and then proceed from their answers.

Time: **Part 2**	**Lesson:** <u>Chapter 1.1 Power Function</u> Give an overall example of the form polynomial function takes, and then identify odd and even functions. Determine what makes a function odd and what make it even. For all even and odd functions, show through graphing each polynomial to better understand similarities and differences. Continuing the use of graphs, show the domain, range, end behaviours and symmetries of different polynomial functions. <u>Chapter 1.2 Characteristic of Polynomial Functions</u> For the two categories of functions (odd and even), show the transformation of the graphs of a function when its coefficient is negative. Identify the similarities and differences between these four groups (even positive, even negative, odd positive and odd negative). Conclude each group's features. Determine the significance of a leading coefficient and the degree of the polynomial in relation to its local minimum and maximums. <u>Chapter 1.3 Equation and Graphs of Polynomial Functions</u> Make a connection between the equations and graphs of polynomial functions. Establish how to graph a function when given an equation and vice versa. Remember to include and mark critical points within all graphs. It is crucial that students will know how to graph polynomial equations using factored form, e.g. where the zeros are (x-intercepts) and the degree of those zeros, hence how the graph behaves when coming in contact with those zeros.
Time: **Part 3**	**Wrap-up:** Ask students to demonstrate their acquired knowledge by applying them to example problems and then answer any unresolved questions.

Notes, reminders & homework:

<u>Reminder</u>:

- Quiz #1 (to be written before / after lesson #2)
- Test #1 (to be written before / after lesson #3)

<u>Homework</u>:
Chapter 1.1: #1 – 6, 9, 10
Chapter 1.2: #1 – 9, 11
Chapter 1.3: #1 – 5, 7, 8, 10, 11

Lesson Reflection: